

Question Paper

Exam Date & Time: 30-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

SUBJECT: ECE 2122/ECE_2122 NETWORK ANALYSIS

(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

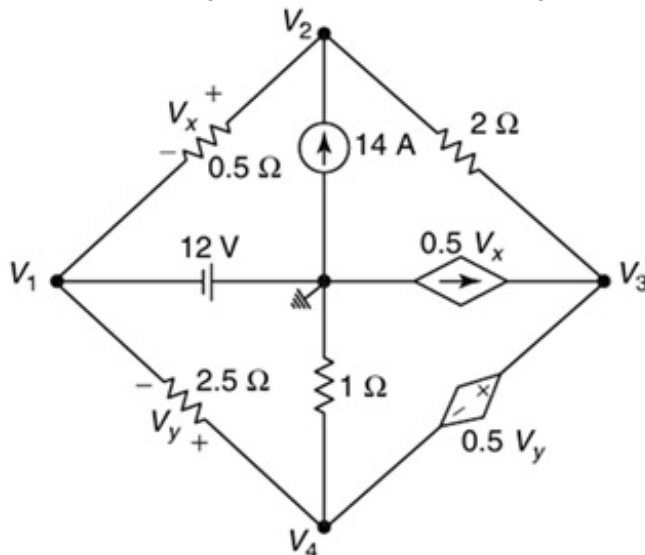
Marks: 50

Duration: 180 mins.

Answer all the questions.

1A) Find the node voltages in the network shown in Figure.

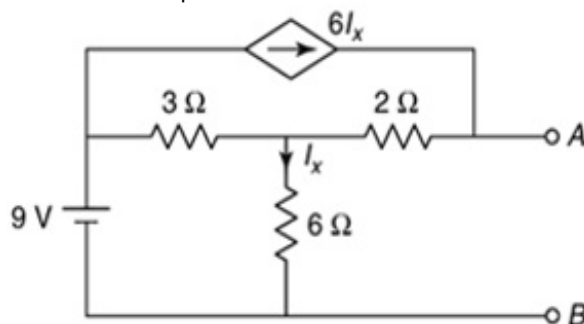
(4)



Course Outcome: "CO1" / Cognitive Level: "Application"

1B) Find Norton's equivalent network at the terminals A and B of Figure.

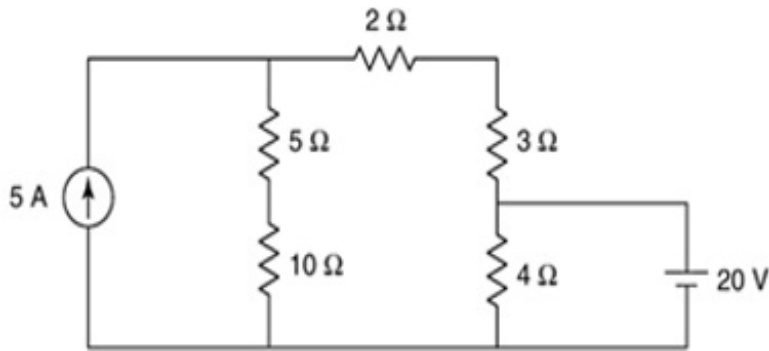
(4)



Course Outcome: "CO1" / Cognitive Level: "Application"

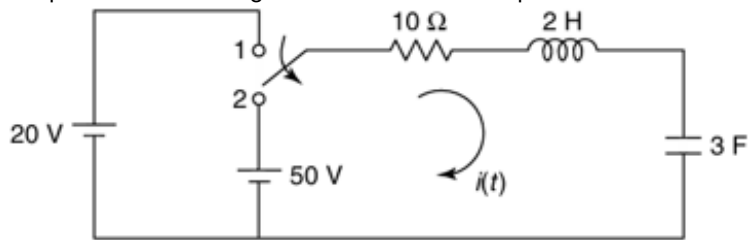
1C) Find the current through the 3 Ohms resistor in Figure using superposition theorem.

(2)



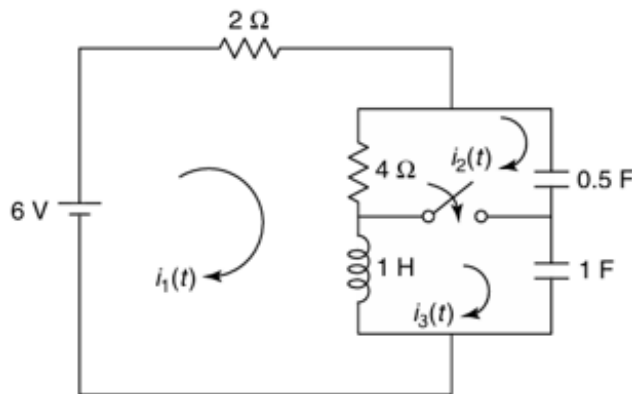
Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) In the network shown in Figure, the switch is moved from the position 1 to 2 at $t = 0$. The switch is in the position 1 for a long time. Determine the expression for the current $i(t)$. (4)



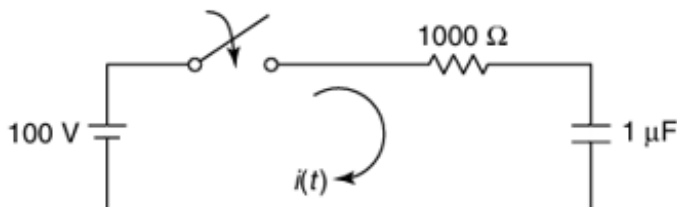
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) In the network shown in Figure, a steady state is reached with switch open. At $t = 0$, switch is closed. Find the three loop currents at $t = 0^+$. (4)



Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) In the network of Figure, the switch is closed at $t = 0$. With the capacitor uncharged, find the value for i , and $\frac{di}{dt}$ at $t = 0^+$. (2)



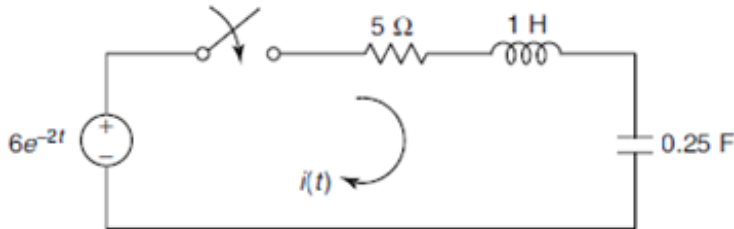
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) If a symmetrical square wave of 4KHz is applied to an RC high-pass circuit and the resultant waveform measured on a CRO was tilted from 14V to 9V, determine the lower 3-dB frequency of the high pass circuit and sketch the waveforms. (4)

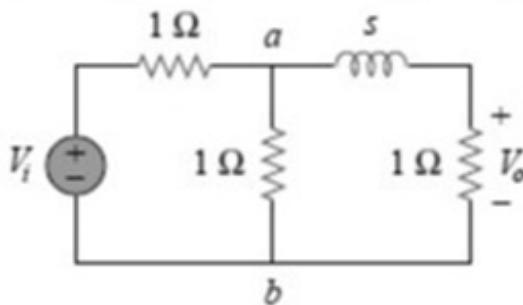
- 3B) A step input of 10 V when applied to the low pass RC circuit produces the output with a rise time of 200 μsec . Calculate the upper 3-dB frequency of the circuit. If the circuit uses a capacitor of 0.47 μF , determine the value of the resistance. (4)

- 3C) Explain under what conditions a high-pass RC circuit can function as a differentiator. (2)

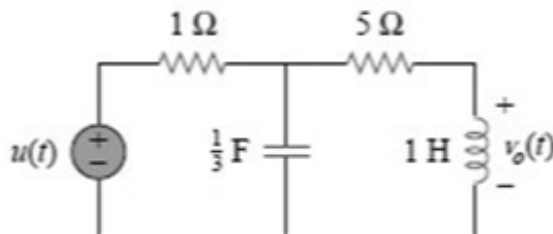
- 4A) In the network shown in Figure, the switch is closed at $t = 0$. Determine the current $i(t)$ using Laplace transform assuming zero initial conditions in the network elements. (4)



- 4B) For the s-domain circuit in Figure, Determine (a) the transfer function $H(s) = \frac{v_o}{v_i}$, (b) the output response when $v_i(t) = u(t)$. (4)

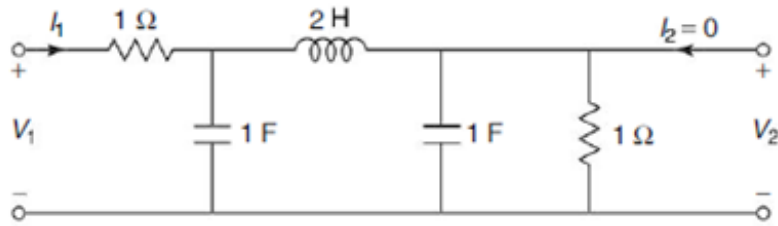


- 4C) Find $v_o(t)$ using Laplace transform in the circuit shown in Figure, assuming zero initial conditions. (2)



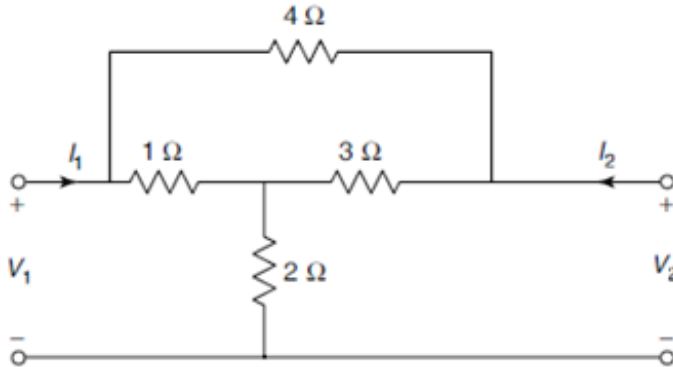
- 5A) (4)

Determine the voltage ratio $\frac{V_2}{V_1}$ for the network shown in Figure.



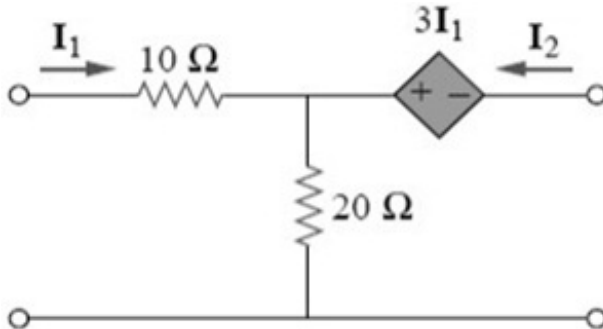
Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Calculate the open-circuit impedance parameters for the network shown in Figure. Determine whether the given network is symmetrical and reciprocal. (4)



Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Determine the transmission parameters for the network in Figure. (2)



Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_2122:NETWORK ANALYSIS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	14	48(96%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)